

SOLAR POWER GENERATION PREDICTION

USING MACHINE LEARNING

TEAM NO-24

Sunidhi Bhakuni - ExecutionLead / 2026DAB1308

Satinderpal Singh - Documentation Lead / 2026EEB1408

Mallarapu Jeswanth Sai - PPT Lead / 2026MCB1542

Sarina - Video Lead / 2026AIB1075

GITHUB REPOSITORY

https://github.com/sunidhibhakuni29-max/Solar_Power_Output

PROBLEM AND DATASET

THE CHALLENGE

Solar output changes with weather conditions, making generation hard to predict.

WHY IT MATTERS

Better forecasts support grid balancing, energy planning, storage decisions, and cost savings.

DATASET

Source	Kaggle dataset by pythonafroz
Size	Approximately 8760 rows (24 hours × 365 days)
Features	Six weather features plus hour of day
Target	Power Production

APPROACH

WORKFLOW

STEP 1
Data Collection

→

STEP 2
Data Preprocessing

→

STEP 3
Train/Test Split

→

STEP 4
Linear Regression Model

→

STEP 5
Prediction and Evaluation

TOOLS & LIBRARIES

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

RESULTS

MODEL PERFORMANCE COMPARISON

Evaluation of Linear Regression and Random Forest models

METRIC	LINEAR REGRESSION	RANDOM FOREST
MAE	249.2116	112.8519
MSE	Not provided	Not provided
RMSE	348.9186	302.9393
R^2	-0.2629	0.0480

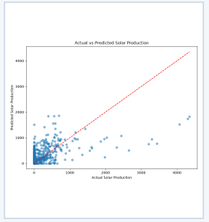
KEY TAKEAWAY

Random Forest has lower MAE and RMSE, and a higher R^2 than Linear Regression in the reported results.

KEY ANALYSES AND VISUALIZATIONS

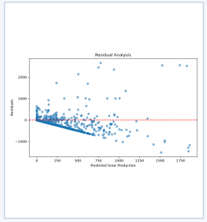
MODEL PERFORMANCE AND NOTEBOOK OUTPUTS

1 ACTUAL VS PREDICTED (SAMPLE ROWS)



Notebook chart: Actual vs Predicted Solar Production.

2 RESIDUAL ERROR (SAMPLE ROWS)



Notebook chart: Residual Analysis.

3 FEATURE INPUTS

Six model input features

- Radiation
- Sunshine
- AirTemperature
- WindSpeed
- AirPressure
- RelativeAirHumidity

Target: SystemProduction

4 MODEL ERROR COMPARISON



Notebook chart: Training vs Testing RMSE.

5 JUPYTER NOTEBOOK OUTPUT

Model evaluation

MAE	143.705
MSE	79844.269
RMSE	282.567
R^2	0.1718

First 5 sample predictions

Row	Actual	Pred.	Error
0	0.00	50.86	-50.86
1	0.00	28.29	-28.29
2	0.00	32.82	-32.82
3	0.00	14.96	-14.96
4	0.00	0.00	0.00

Training RMSE: 988.898
Testing RMSE: 282.567

LEARNINGS

WHAT WORKED

Weather features such as irradiance and hour of day gave a strong, interpretable baseline with a simple model.

WHAT DID NOT

Linear models miss non-linear day-night and cloud effects. Errors are highest at sunrise, sunset and cloudy hours.

DIFFICULTIES

Understanding the dataset well took time.
Not being well familiar with Git and GitHub was a challenge initially.

TEAM CONTRIBUTIONS

- Sunidhi — model code and experiments
- Satinderpal — project documentation
- Mallarapu — presentation slides
- Sarina — project video

AI DISCLOSURE AI was used to assist in preparing and designing this presentation.